

BookLoop - Capstone Presentation Walkthrough

1. INTRODUCTION (1-2 mins)

Speaker:

"Good morning everyone. Today we are proud to present BookLoop - a smart, sustainable peer-to-peer book exchange platform built specifically for college campuses.

The problem we are solving is the high cost of academic textbooks and the inefficiency of buying and selling them locally. Our solution introduces a virtual campus currency called 'BookCoins' to facilitate a cashless economy, powered by three distinct AI and Machine Learning models."

2. ARCHITECTURE OVERVIEW (1 min)

Speaker:

"Before we jump into the demo, here is a quick overview of our tech stack:

- Our frontend is built with React and Vite.
- Our backend and database are powered by Supabase.
- Our machine learning microservice is built with FastAPI, Python, and PyTorch.
- We utilize Google Gemini Vision for AI image appraisal, Fuzzy Logic for dynamic valuation, and an Artificial Neural Network (ANN) for personalized recommendations."

3. DEMO: LISTING A BOOK & AI APPRAISAL (2-3 mins)

[ACTION: Log in to the Seller account and click 'List Book']

Speaker:

"Let's list a book on the platform. We fill in the details and our self-assessment of the condition. But to prevent fraud, we don't just take the seller's word for it."

[ACTION: Upload photos and click submit to trigger the AI Review]

Speaker:

"Once submitted, these photos are sent to our Google Gemini Vision AI via a Supabase Edge Function. The AI acts as a digital inspector, identifying physical damage and verifying our claim. It then passes a condition score to our Fuzzy Logic Engine, which dynamically calculates a fair BookCoin price based on original MRP, demand, and condition."

4. DEMO: THE BUYER EXPERIENCE & ANN RECOMMENDATIONS (2 mins)

[ACTION: Log out of Seller, log into Buyer account (demo@demo.com)]

Speaker:

"Now, switching to a buyer's perspective. When we land on the Dashboard, you'll immediately see our 'AI-Powered ANN Recommendations' section."

[ACTION: Point out the Neural Network badge on the dashboard]

Speaker:

"Instead of a generic list of books, our FastAPI ML service runs a PyTorch Artificial Neural Network. It takes a 10-dimensional feature vector - including the user's campus preferences, price history, and book categories - to rank and serve the highest probability matches specifically for this user."

5. EXPLAINING THE TRUST SCORE SYSTEM (1 min)

Speaker:

"To maintain platform integrity, we also developed a Trust Score System using Fuzzy Logic. After every transaction, buyers rate the book's condition. If a seller repeatedly uploads misleading photos, our ML model automatically recalculates and tanks their Trust Score, effectively penalizing bad actors in our campus economy."

6. CONCLUSION (1 min)

Speaker:

"In conclusion, BookLoop successfully integrates Generative AI and classic Machine Learning to solve real-world marketplace fraud and pricing issues, creating a sustainable and smart campus economy. Thank you. We are now open for questions."